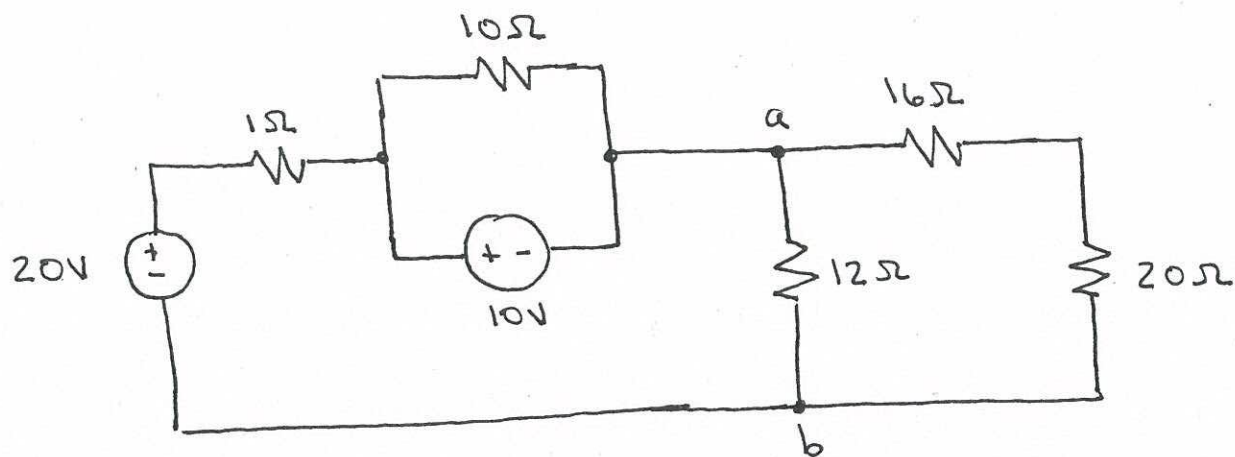
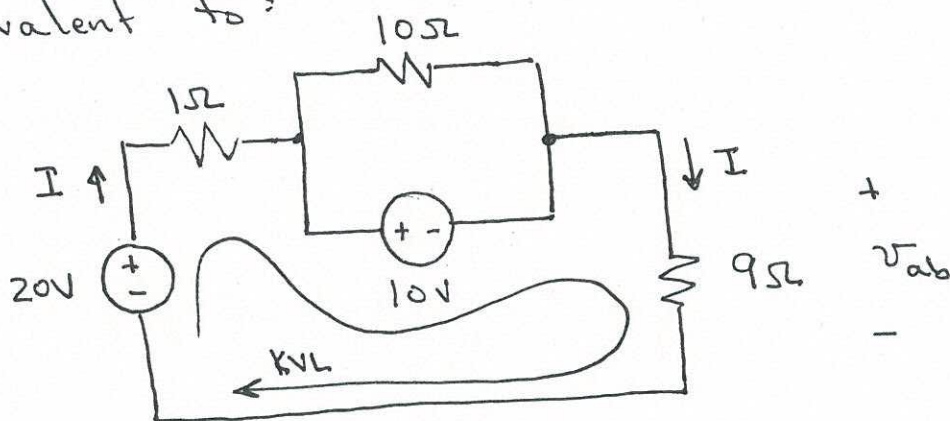


2.1



Using circuit reduction, the above circuit is equivalent to:



$$\text{KVL, Loop shown: } -20\text{V} + 1\Omega(I) + 10\text{V} + 9\Omega(I) = 0$$

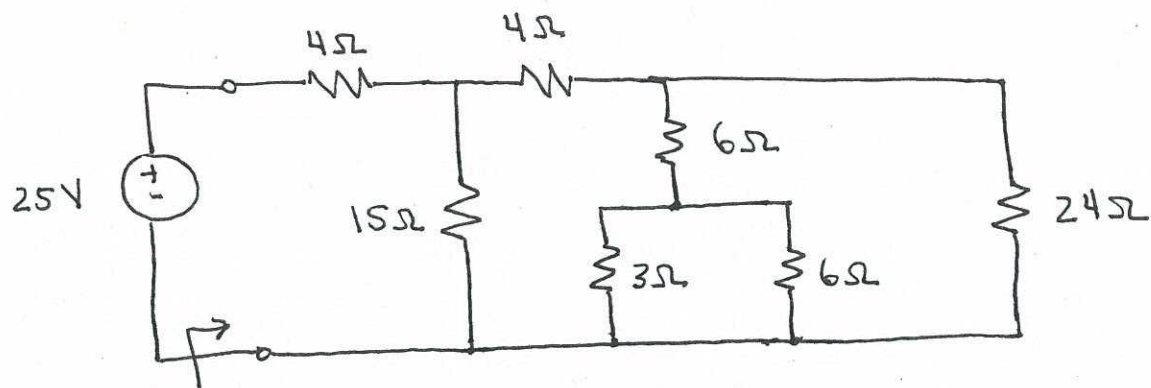
$$I = 1\text{A}$$

Ohm's Law:

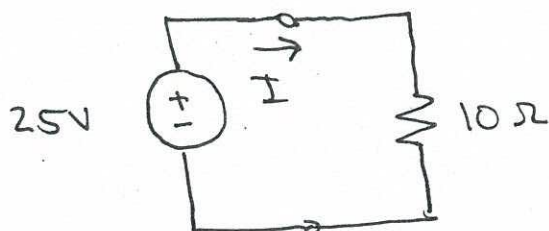
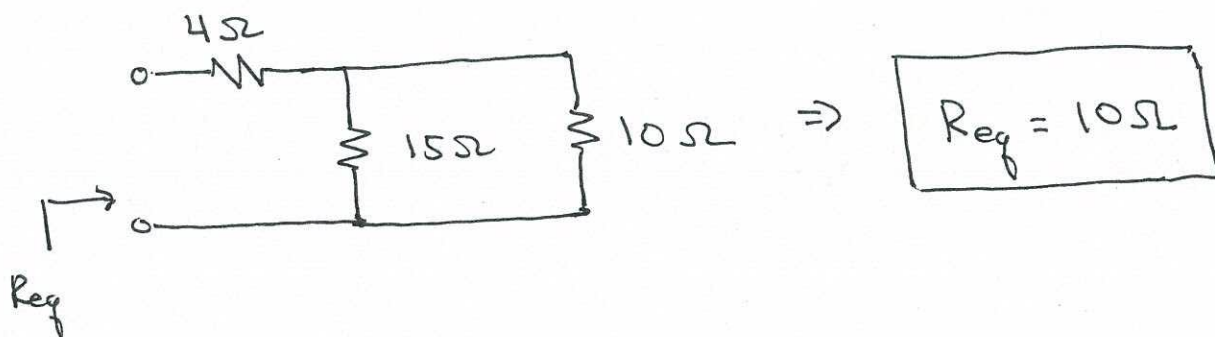
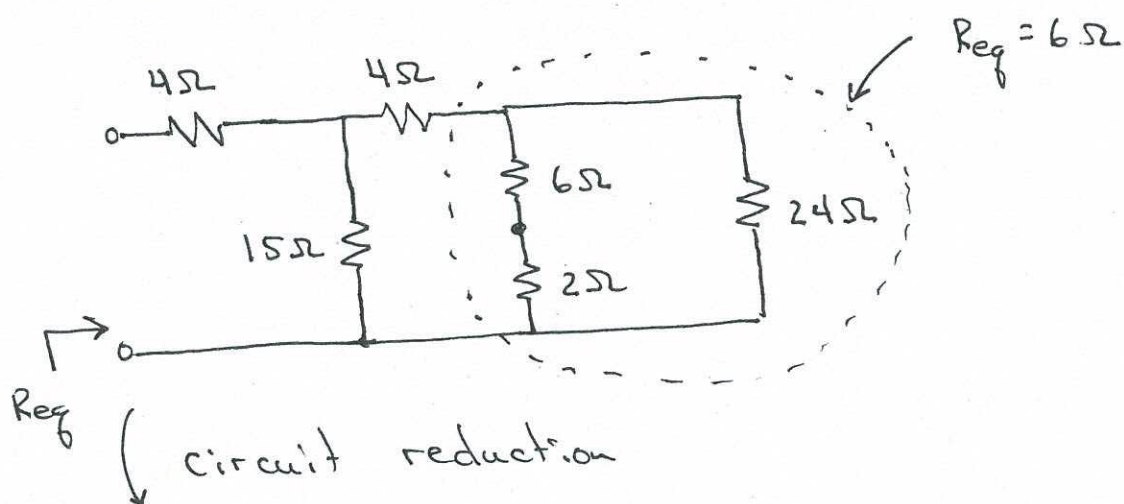
$$v_{ab} = (9\Omega)(1\text{A}) = \boxed{9\text{V} = v_{ab}}$$

$$P_{20\text{V}} = I(20\text{V}) = (20\text{V})(1\text{A}) = \boxed{20\text{W (generated)}}$$

2.2



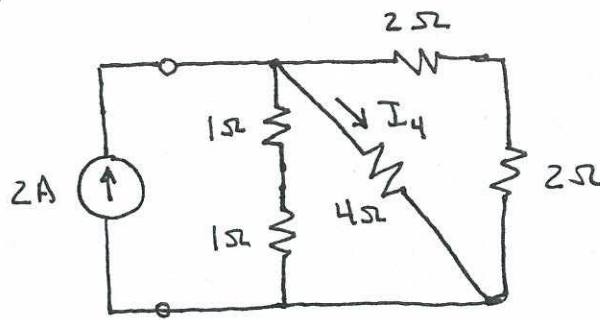
R_{eq} $\downarrow$ circuit reduction



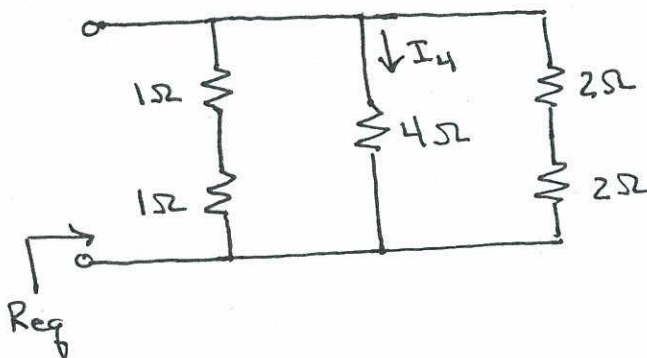
$$I = \frac{25V}{10\Omega} \text{ (ohm's law)}$$

$$I = 2.5A$$

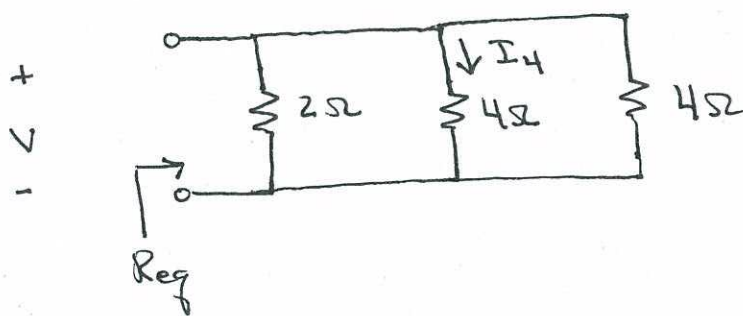
2.3



Re-draw circuit to emphasize parallel, series combinations

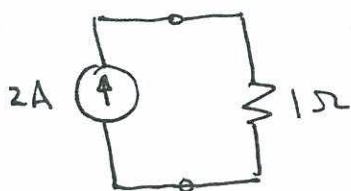


Combine series resistors:



$$\Rightarrow R_{eq} = \frac{1}{\frac{1}{2} + \frac{1}{4} + \frac{1}{4}} = 1\Omega$$

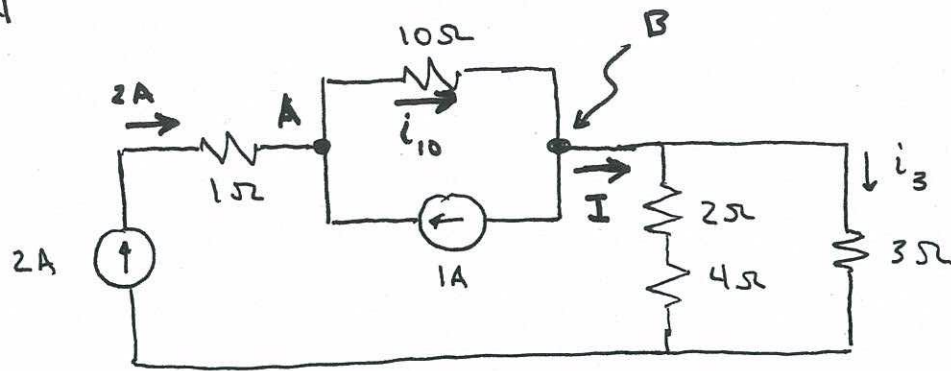
Reduced circuit:



$$+ V = 2V = (2A)(1\Omega) \Rightarrow I_4 = \frac{2V}{4\Omega}$$

$$I_4 = 0.5A$$

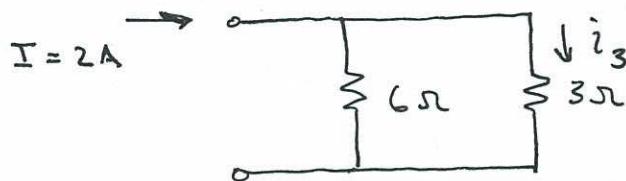
2.4



$$\text{KCL at A: } 2\text{A} + 1\text{A} - i_{10} = 0 \Rightarrow i_{10} = 3\text{A}$$

$$\text{KCL at B: } i_{10} - 1\text{A} - I = 0 \Rightarrow I = 2\text{A}$$

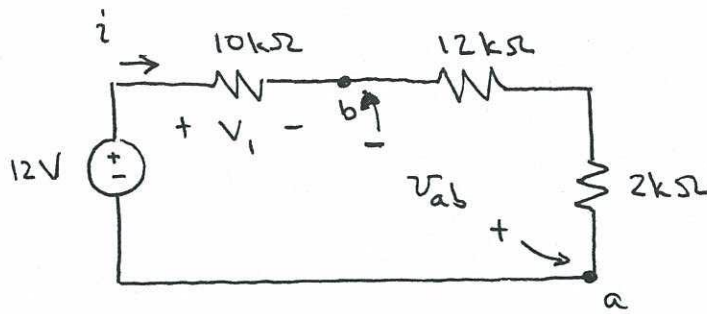
Combine the series combination of the 2Ω and 4Ω resistors to obtain the current divider below:



$$i_3 = 2\text{A} \left(\frac{6\Omega}{3\Omega + 6\Omega} \right)$$

$$\underline{\underline{i_3 = \frac{4}{3}\text{A}}}$$

2.5



$$\text{KVL: } -12\text{V} + (10\text{k}\Omega)i + (12\text{k}\Omega)i + (2\text{k}\Omega)i = 0$$

Note: currents through all elements are the same, by KCL.

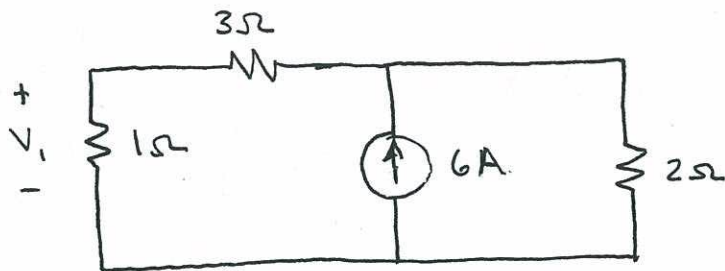
$$i = \frac{12\text{V}}{10\text{k}\Omega + 12\text{k}\Omega + 2\text{k}\Omega} \Rightarrow i = 5\text{mA}$$

$$V_1 = (10\text{k}\Omega)i = (10\text{k}\Omega)(5\text{mA}) \Rightarrow \underline{\underline{V_1 = 5\text{V}}}$$

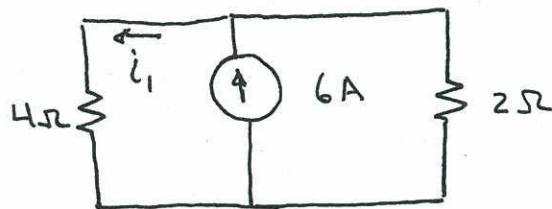
$$\text{KVL: } -12\text{V} + V_1 - v_{ab} = 0$$

$$-12\text{V} + 5\text{V} - v_{ab} = 0 \Rightarrow \underline{\underline{v_{ab} = -7\text{V}}}$$

2.6



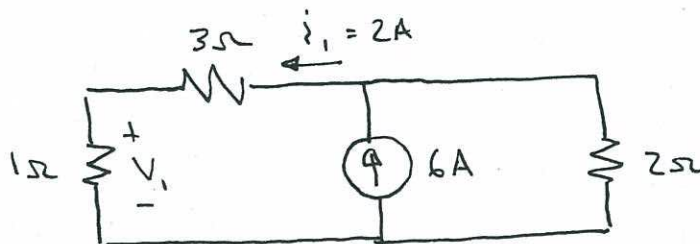
Combine 1Ω & 3Ω resistors:



The above is a current divider, so:

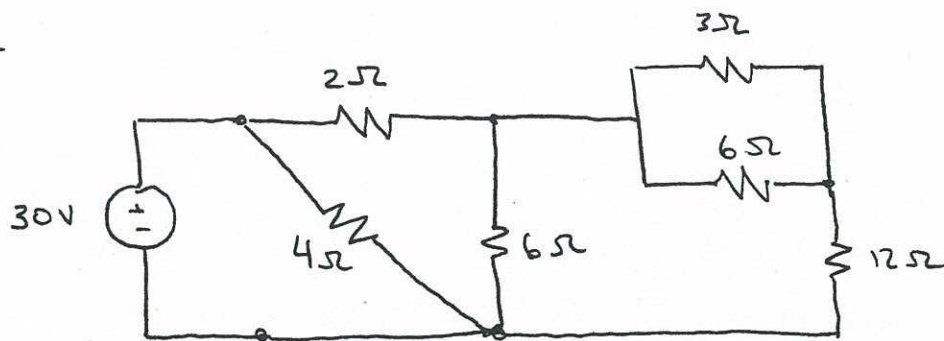
$$i_1 = 6A \left(\frac{2\Omega}{2\Omega + 4\Omega} \right) = 2A$$

Now, from original circuit:

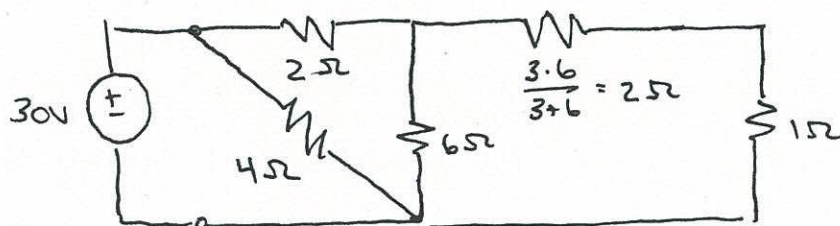


$$V_1 = (1\Omega)(2A) \Rightarrow \underline{\underline{V_1 = 2V}}$$

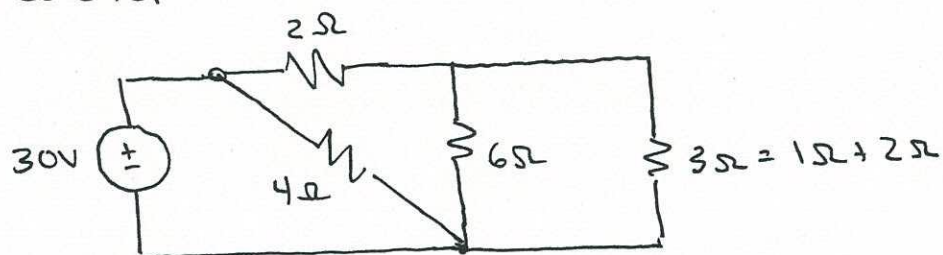
2.7



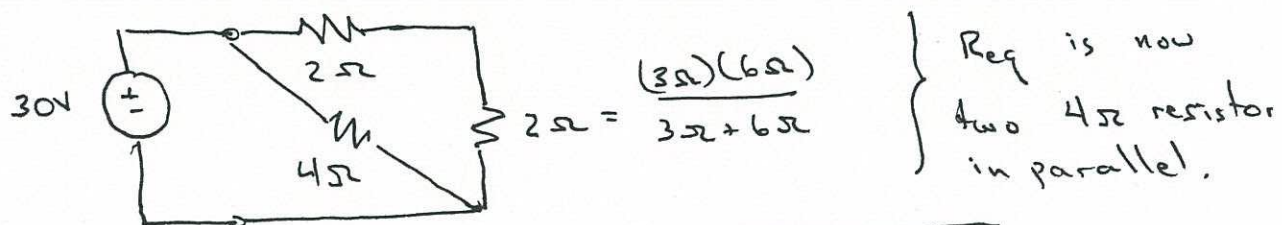
(a) First, combine parallel 3Ω & 6Ω resistors:



The resulting 2Ω resistor is in series with 1Ω resistor:



The 3Ω & 6Ω resistors are in parallel:



The equivalent resistance, $\boxed{R_{eq} = 2\Omega}$

(b) $I = \frac{30V}{R_{eq}} = 15A \Rightarrow P_s = (-30V)(15A)$

$\boxed{P_s = -450W}$ (generated)

2.7 (cont'd)

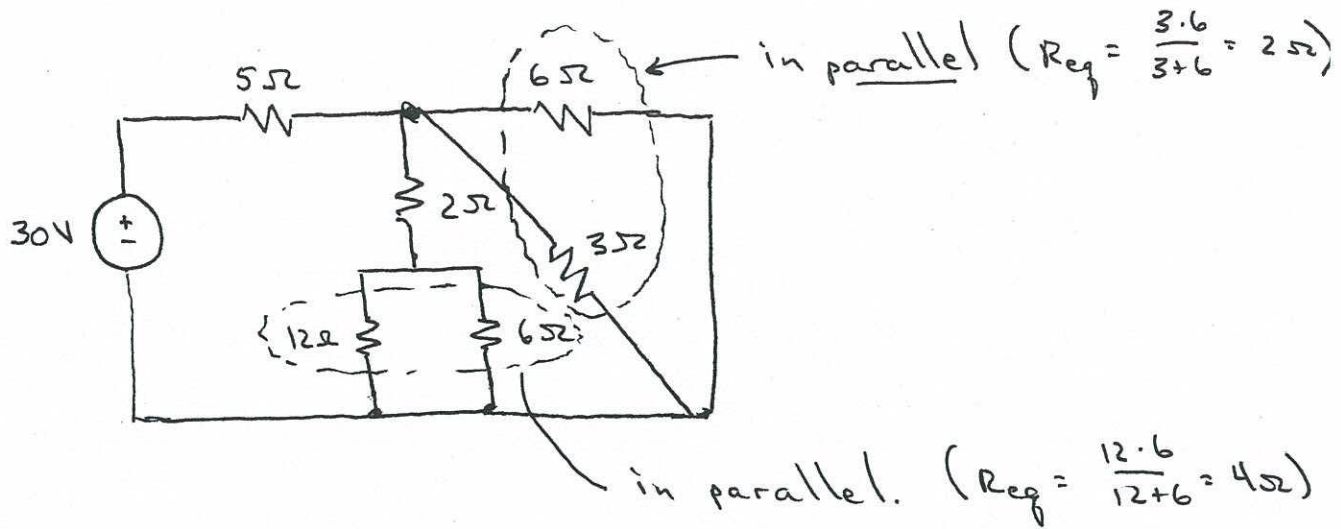
(c) The current through the 4Ω resistor can be found directly from Ohm's Law. The voltage across the 4Ω resistor is

30V. Therefore,

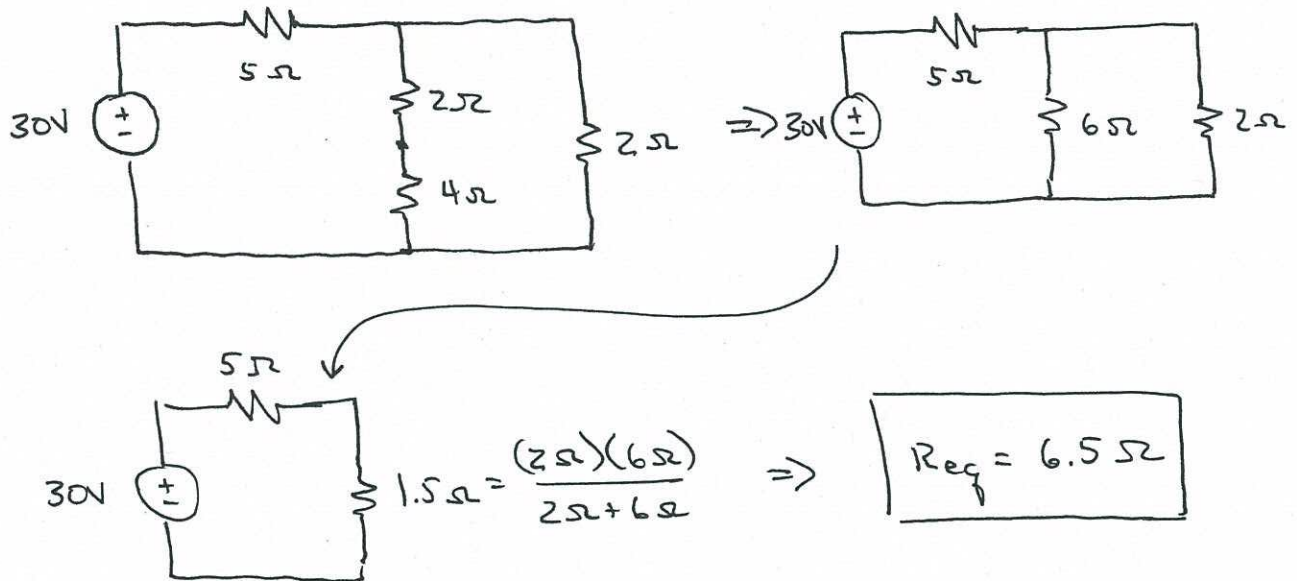
$$I = \frac{30V}{4\Omega} = 7.5A$$

2.8

(a)

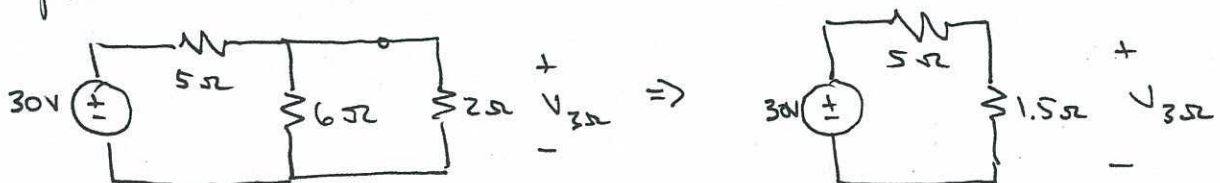


Combining the two sets of parallel resistors:



(b) $I_s = \frac{V_s}{R_{eq}} \Rightarrow I_s = \frac{30V}{6.5\Omega} \Rightarrow \boxed{I_s = 4.615 A}$

(c) The voltage across the 3Ω resistor is the same as the voltage across the parallel combination of the 3Ω & 6Ω resistors. We can use the previous circuit reductions:



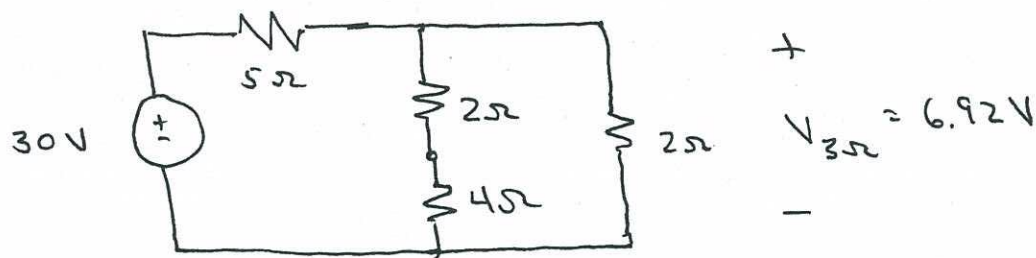
2.8 (cont'd)

The voltage across the 3Ω resistor can now be found from a voltage divide formula:

$$V_{3\Omega} = 30V \left[\frac{1.5\Omega}{5\Omega + 1.5\Omega} \right]$$

$$V_{3\Omega} = 6.92V$$

(d) We can again take advantage of an earlier circuit reduction.

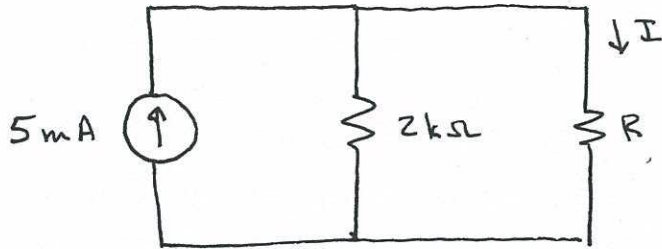


The 2Ω and 4Ω resistors now form a voltage divider. The total voltage across the series combination is $6.92V$. Therefore,

$$V_{2\Omega} = 6.92V \left[\frac{2\Omega}{2\Omega + 4\Omega} \right]$$

$$V_{2\Omega} = 2.31V$$

2.9



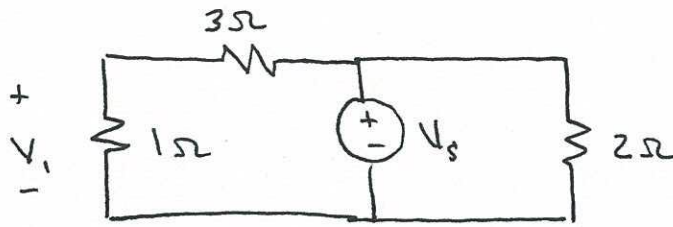
current divider: $I = 5\text{mA} \left[\frac{2\text{k}\Omega}{2\text{k}\Omega + R} \right] = 2\text{mA}$

$$5\text{mA} [2\text{k}\Omega] = 2\text{mA} (2\text{k}\Omega + R)$$

$$\frac{5}{2} (2\text{k}\Omega) = 2\text{k}\Omega + R$$

$$\underline{\underline{R = 3\text{k}\Omega}}$$

2.10



Voltage divider : $V_1 = V_s \left[\frac{1\Omega}{1\Omega + 3\Omega} \right] = 2V$

$$\underline{\underline{V_s = 8V}}$$